B-B-Bugging

AGENTVERSE 2026 · FINANCE TRACK

Campus Index

Two cooperating agents that coach UK students on paper ETF investing, and check the rent is safe before looking at a single price.

> Tested on 30 unseen markets: a smaller worst fall than 60/40 in all 30, and a 57% smaller worst-case loss (−8.5% against −19.9%).

Research prototype · Fictional students and holdings · Synthetic prices · Not financial advice

THE USER AND THE FAILURE

A student about to invest their rent

> *“My £1,500 is for rent next term and I have about 1 month saved. My mate says go all in on Nasdaq for 100% a year.”*
>
> Fictional student · the app’s “Rent at risk” preset

Money that is already spoken for

The £1,500 has to pay next term’s rent.

No cushion

One month of cash runway, so any fall becomes missed rent.

A target from a friend

100% a year, concentrated in one market, decided in minutes.

WHAT WE BUILT

Three gates before anything changes

01

Affordability veto

Rules check rent exposure, cash runway, horizon and loss limit before any prices load.

02

Risk Guardian

A model-free uAgent reruns the analysis, matches the digest and rejects any figure it cannot recompute.

03

Human approval

Only the student can apply a change to the paper ledger. The agents have no write tools.

The model chooses tools and writes text. It never publishes a number the system hasn’t verified.

HOW IT WORKS

The maths never touches the model

The student describes their situation in plain English →

LLM

Rent-safe Coach

Claude, ASI:One or an offline planner. Quotes the student, asks what's missing, picks tools. Max 8 calls.

→

NO MODEL

Verified engine

Parse, normalise, enrich, analyse, verify, explain. Veto, ranking, stress tests and limits.

→

NO MODEL

Risk Guardian

Separate uAgent. Reruns everything, matches the digest, checks every £, % and × figure.

→

HUMAN

Human approval

The student reviews and approves. Only then does the paper ledger change.

`audit` Append-only log of every tool call, quote and Guardian verdict.

coach.py · planner.py · agent.py · strategy.py · verifier.py · guardian.py · paper.py · agents_app.py (uAgents)

WHY IT'S AN AGENT

Each student gets a different plan of action

RENT AT RISK

```
1 update_profile · 4 quoted facts
2 run_analysis · veto
3 get_evidence
→ declines, explains why
```

SUMMER JOB SAVINGS

```
1 update_profile · 5 quoted facts
2 run_analysis · ready
3 what_if · 6 months' cash first
4 get_evidence
→ 12 cited figures, countersigned
```

Asks, doesn't assume. If rent, cash runway or horizon is missing, the Coach asks the student first.

Plans its own steps. It picks tools and what-ifs to fit the student, up to 8 calls.

Checked by a second agent. The Guardian countersigns or sends the answer back.

Lives on Agentverse. Both are uAgents; the Coach speaks the chat protocol for ASI:One.

Real tool sequences, 8 Nov 2026 · The same loop runs on Claude or ASI:One; the offline planner keeps the demo deterministic.

THE COACH EXPLORES ON ITS OWN

It asks what it needs, then tries an alternative

Student: “I’ve got £1,200 from my summer job that I won’t need for rent or bills. I won’t touch it for 3 years. Can I put it in ETFs?”

Coach asks: “Before I analyse anything: how many months of essential costs do you hold in cash outside this plan?”

Student: “I only have about 1 month of costs saved.”

```
run_analysis · veto: reserve under 3 months
what_if · 3 months’ cash first: ready
```

	Original plan	Coach’s what-if
Cash reserve	1 month	3 months
Result	Blocked	Paper allocation allowed
Allocation	None	EMIM 20.6%, VGOV 20.6%, VEUR 17.6%, IGLS 5.4%, rest in cash

Coach’s answer: not yet, because the reserve is under three months; with three months’ cash first, a paper allocation becomes possible. The Guardian countersigned it.

Real two-turn conversation, fictional student, 8 Nov 2026 · The same loop runs on Claude or ASI:One; the offline planner keeps the demo deterministic.

INSIDE THE TWO MODEL-FREE PARTS

The engine decides. The Guardian checks.

Verified engine

```
score = 0.55·m60 + 0.35·m20 + 0.10·m5
− 0.20·volatility − fund fee − 0.15% cost
```

- Rank 19 UK-listed ETFs by trend; buy equities only when the trend is positive
- Cap each fund and exposure group at 30%; skip near-duplicates
- Shrink the portfolio until the worst of three stress tests fits the student's loss limit
- Risk level 1–5 caps total and equity exposure

Risk Guardian

Trusts none of the Coach's numbers. Rejects unless all four checks pass:

1. Affordability rules re-applied: a veto must stand
2. Whole decision rerun independently: digest must match
3. Only verified, unlevered allocations reach approval
4. Every £, % and × in the Coach's text matches a recomputed fact

Fail: sent back once, then replaced by a verified template.

strategy.py · verifier.py · guardian.py · m60, m20, m5 = 60-, 20- and 5-day price momentum

TRACEABLE REASONING

Every figure cites a fact the Guardian recomputed

COACH ANSWER · EXCERPT

The verified allocation is EMIM 20.6%[1], VGOV 20.6%[2], VEUR 17.6%[3], IGLS 5.4%[4], with £714.55 left as cash[5]. The worst modelled shock would lose 19.7%[6] against your 20.0% loss limit[7]. Treat it as a way to limit losses, not to chase returns.

`[1] base.weight.EMIM · [5] base.cash · [6] base.stress.joint_drawdown`

RISK GUARDIAN · COUNTERSIGNED

- Affordability rules re-applied
- Independent rerun matched the proposal digest
- Unlevered and verifier-passed
- Every figure matches a cited, recomputed fact

Student's words quoted into the profile: "4 months", "3 years", "£2,000".

Real output, "Summer job savings" preset, 8 Nov 2026, synthetic prices (seed 261107) · The same loop runs on Claude or ASI:One; the offline planner keeps the demo deterministic.

THE FAILURE, HANDLED ON SCREEN

Rent at risk: it stops before loading prices

ENGINE TRACE

```
✓ parse · profile validated
✓ normalise · register checked
× enrich · essential expenses veto
– market, analyse, verify · not run
```

Quoted facts: “is for rent”, “1 month”, “£1,500”, “100% a year”.

> “Investing this money is not appropriate yet: the fictional funds may be needed for rent, tuition or living expenses. The Risk Guardian will not let a proposal through until that rule is met.”

Under test, a reckless model answers “Put 100% in CNX1”. The Guardian sends it back, swaps in a verified template, and the veto holds.

Real output, “Rent at risk” preset, 8 Nov 2026 · test_rent_veto_survives_a_reckless_model · The same loop runs on Claude or ASI:One; the offline planner keeps the demo deterministic.

DATA

Synthetic, labelled and licence-checked

Source	What we use	Rights
Our simulated GBP prices	956 observations per market, seeded equity and rate regimes	Authored by us. No third-party prices.
Vanguard UK, iShares UK	19 LSE GBP UCITS listings: ISIN, domicile, fee, benchmark	Factual fields and issuer links only
Agentverse finance kit	Hidden 60-case markets challenge	Our results only; no fixtures or answer keys shipped
Multi-seed evaluation	30 fresh seeds, hash-committed before generation	Authored synthetic paths

No real ETF prices were used. Yahoo data was left out because its terms restrict automated collection. A licensed price file can be loaded, with source, date and licence, when one is available.

TESTING ON UNSEEN DATA

We committed to the test before the markets existed

1

Freeze the rules

Strategy files hashed. Tuning only on disclosed design seeds 1–10.

2

Commit the seeds

30 fresh seeds drawn; their SHA-256 written down before any path existed.

3

Generate 30 markets

Each with its own noise and a shuffled order of eight drift regimes.

4

Replay with costs

756 observations after a 200-observation warm-up, against two benchmarks.

Benchmarks

A 60/40 portfolio, and the student's own limits held statically with no signals, to isolate what the timing adds.

Real costs

Next-close fills, 10 bp adverse half-spread, 5 bp commission, fund charges accrued daily.

python3 backtest_seeds.py verify recomputes the commitment (re-checked 8 Nov 2026) · evidence/multi-seed/

RESULT · 30 UNSEEN MARKETS

It doesn’t make more money. It reliably loses less.

30/30

markets where Campus Index had a smaller worst fall than a 60/40 portfolio. Sign test p < 0.001: not luck.

8.7%

median cut in worst fall vs 60/40, in percentage points

57%

smaller worst-case loss: −8.5% in its worst market, against −19.9% for 60/40

5.2%

the cost of protection: median return given up vs 60/40, in percentage points

campus-ro-1.1, Balanced profile · Median return 9.5% vs 16.5%; beat 60/40 on return in 33% · Worst cases from evidence/multi-seed/results.jsonl

THE KIT CHALLENGE, HONESTLY

Our first score looked good for the wrong reason

RUN 1 · HIDDEN, 60 PATHS

Traded on 1 of 60

Kit paths are 12–78 bars, but our 60-day signal needed 61. The 39–21 record against buy-and-hold was cash sitting out falling markets.

RUN 2 · FRESH HIDDEN RERUN

Traded on 25 of 60

A labelled short-history signal from 21 bars, fixed before the rerun. Beat cash on 10 paths, up from 1. Still 39–21 against buy-and-hold.

It avoided all 20 falling and flat paths and lost all 10 rising ones by entering late. That is the same trade-off as our main result: fewer falls, slower gains.

evidence/kit-challenge/ · evidence/kit-challenge-v2/ · The kit measures single-asset signals only, not the portfolio or suitability workflow

RISK CONTROLS AND OVERSIGHT

Limits live in code, not in the prompt

- **Affordability veto** before any prices load. Shown live.
- **Figure check:** any ungrounded £, % or × is sent back once, then replaced.
- **Position caps:** 30% hard cap per fund and group; standard portfolio at 1.0×.
- **Tool budget** of eight calls; pausing paper actions stops the coach.
- **Digest match:** the Guardian's rerun must equal the proposal.
- **No write tools** for the model. The student approves every change.
- **Leverage** needs explicit acknowledgement and never reaches approval.
- **Append-only audit** of every tool call, quote and verdict.

30 unit tests pass, re-run 8 Nov 2026. They include a rent veto that survives a reckless model, rejection of wrong or uncited numbers, rejection of tampered or levered proposals, and the tool budget.

tests/test_coach.py · tests/test_campus.py

LIMITS AND NEXT STEPS

What this does not show yet

Limits

- Every market is synthetic; no real ETF performance
- The seed commitment is ours, not sealed by the organisers
- No real students have used it yet
- Wrapper eligibility for some funds, holdings overlap and liquidity are unresolved

Stronger evidence would need

- A licensed, point-in-time ETF price panel
- Real execution-cost estimates
- An organiser-sealed holdout
- Testing with real students on fictional money

> **Who it’s for:** student money that might be needed in the next few years, where a big fall would hurt. For long-horizon savings, a plain global tracker may suit better.

Veto. Guardian. Human.

Campus Index never publishes a number it hasn’t verified, and never moves money a student needs for rent.

> A smaller worst fall than 60/40 in 30 of 30 unseen markets ($p < 0.001$). A 57% smaller worst-case loss (−8.5% against −19.9%).

```
python3 server.py --port 8766 · runs offline, no API key needed
```